

User Documentation

Introduction

Bit&Beam is an intelligent document management system for building-related data. It helps users organize, analyze, and retrieve building documents efficiently.

Key Features

- **Document Upload and Storage:** Upload permits, certificates, reports and any building related documents.
- **Automated Processing:** AI-driven OCR and metadata extraction.
- **Intelligent Search:** Natural language querying for building records.
- **Multi-Tenancy:** Secure access with data segregation based on organizations.

Getting Started

1. **Access the Web UI:** Open your browser and go to amos.b-iq.net.
2. **Login:** Use your organization-specific credentials to log in.
3. **Add Building:** Use Add Building button to add a building and relevant information
4. **Upload Document:** Use Upload button to upload a document
5. **Add Category & Building:** Using the AI system category would be auto detected. User can change the category if necessary and also add a relevant building from the dropdown menu.
6. **Summary:** General information about the app can be also found using the AI chatbot on the upload page
7. **Document View:** Clicking on a specific document user can view a particular document, edit, delete, or download as necessary from the page. A relevant summary also can be found using the chatbot

Using the System

Uploading Documents

1. Navigate to the "Upload" section.
2. Drag and drop files or use the "Choose File" button.
3. Uploaded documents will be automatically processed.

Query Documents from API

1. Use API endpoints (following OpenAPI standards) to interact with Documents.



Example User Flows

User Flow 1: Upload a Document with AI Support

Login → Click "Upload" → Choose File → Auto-detection runs → Select Building → Confirm Category → Submit → Use AI chatbot for summary

User Flow 2: Add a New Building

Login → Go to "Buildings" → Click "Add Building" → Fill form → Save → Ready for document linking

User Flow 3: View, Edit, or Download a Document

Login → Navigate to Document List → Click on a Document → Use toolbar (View/Edit/Delete/Download) → Use AI chatbot for help

Contact

For further assistance, contact your system administrator or refer to the technical documentation.

Technical Design Documentation

1. Project Overview

The Bit&Beam project is an intelligent document management system designed for building-related data. It aims to streamline document workflows using automated classification, metadata extraction, and intelligent search capabilities. The system supports multi-tenancy, ensuring data segregation and access control based on organizations.

2. Architecture

The project follows a modular structure, consisting of frontend, backend, and several services for specific tasks.

3. Key Components

- Frontend: Angular-based UI for user interaction.
 - Located in the [frontend/](#) directory.
 - Uses TypeScript and Angular CLI.
- Backend: C# ASP.NET Core Web API.
 - Located in the [backend/](#) directory.
 - Handles API requests, business logic, and data persistence.
- Database: PostgreSQL with pgai extension.
 - Configuration and schema in [database/](#) directory.
 - Search: Opensearch for indexing and full-text search.
- AI/Extraction:
 - Ollama for AI integration. Located in the [ollama/](#) directory.
 - Apache Tika for metadata extraction. See [TikaService.cs](#).

4. Data Model

The data model is defined using C# classes in the [backend/BUILD.ING/Models/](#) directory. Key entities include:

- User: System users information.
- Building: Stores building details.

- Document: Stores document metadata and file path.
- Organization: Represents a tenant.

5. Key Technologies

Programming Languages: TypeScript, C#, Python

Frameworks/Libraries: Angular, ASP.NET Core

Databases: PostgreSQL

AI systems: Ollama, Apache Tika

Containerization: Docker, Docker Compose

6. Development Guidelines

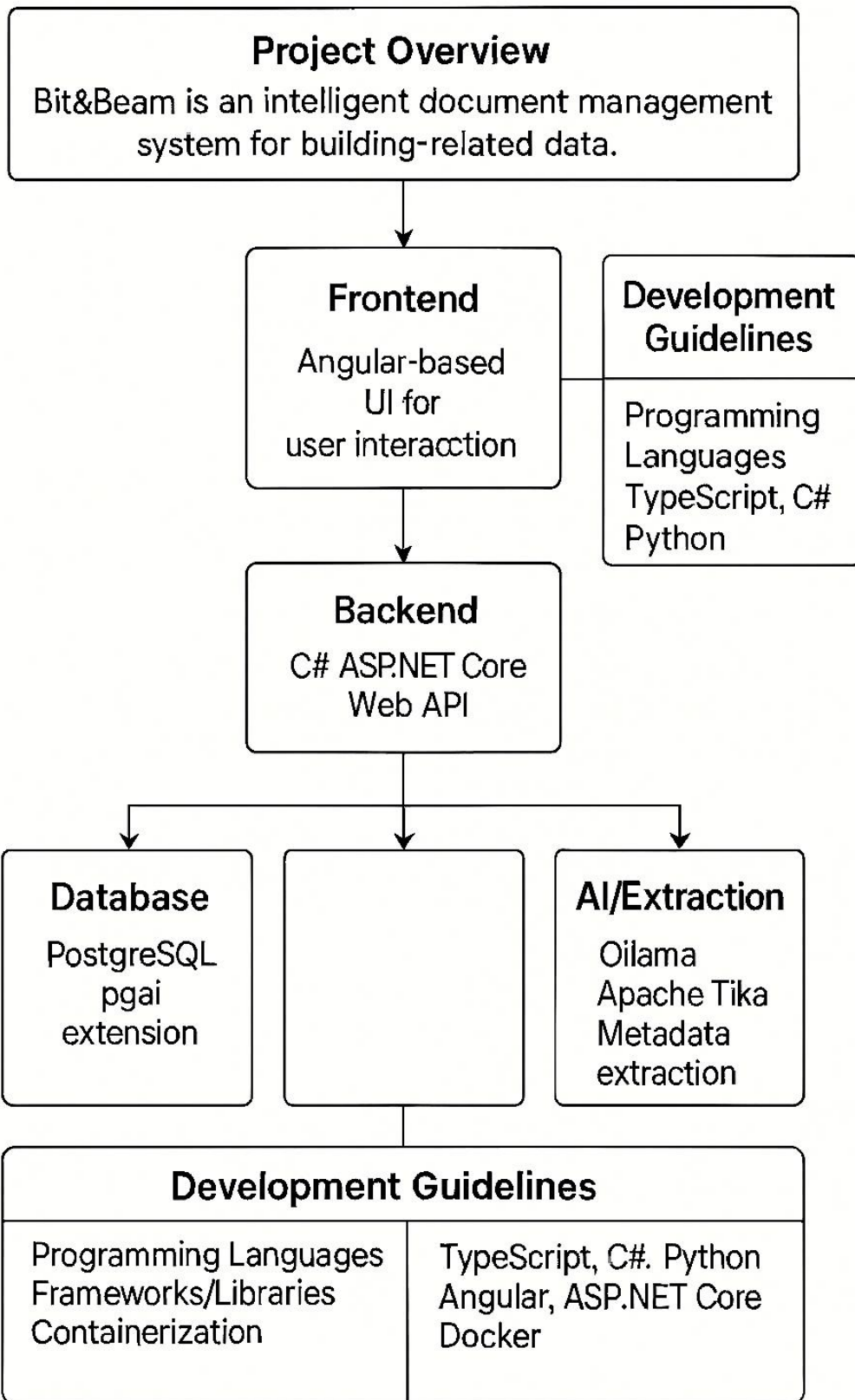
See [Coding_Guidelines.md](#)

- Protected main and dev branches.
- Feature branches.
- Pull requests require review from at least two other developers.

7. Build and Deployment

Docker Compose is used for orchestrating services. See [docker-compose.yml](#) and [docker-compose-prod.yml](#)

GitHub Actions workflows for linting and CI/CD. See [.github/workflows/](#)



Build & Deploy Guide

This project includes a **C# backend** (OpenAPI-based), an **Angular frontend**, and uses **Ollama**, **Apache Tika**, and **PostgreSQL** for various services. The development and deployment environments are fully containerized using Docker.

Prerequisites

- [Docker](#)
 - [Docker Compose](#)
 - [GitHub CLI \(optional\)](#) (GitHub Workflows)
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Getting Started Locally

1. Build & Start All Services

From the /BitAndBeam directory, run:

```
docker-compose up --build
```

This will:

- Build and run the C# backend
- Serve the Angular frontend
- Start Ollama Service
- Launch Apache Tika (document parsing)
- Set up a PostgreSQL database instance

2. Accessing Local Services

- **Frontend:** <http://localhost:8080>
 - **Backend API:** <http://localhost:5001>
 - **PostgreSQL:** localhost:5432 (credentials in .env or docker-compose.yml)
 - **Tika Server:** <http://localhost:9998>
 - **Ollama:** <http://localhost:11434>
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GitHub Actions CI/CD Pipeline

This project includes a **GitHub Actions** pipeline for automated builds and deployments to a server.

What It Does

- Runs on every push to the main branch
- Builds Docker images for each service
- Pushes them to the GitHub Container Registry
- Remotely runs containers on the production server

Workflow File

See `.github/workflows/docker-ci.yml` for the CI/CD definition.